

Jakob Olof Popov Wirén

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Swedish Citizen, Swiss Resident

SUMMARY

ML/AI engineer shipping production LLM agent systems in aerospace. Company-wide AI-integration lead at XER Technologies, with a research background in applied machine learning. First-place finish at the Andon Labs LLM Safety Hackathon. M.Sc. Engineering Physics, Lund University, specialising in Control Theory & Machine Learning.

PROFESSIONAL EXPERIENCE

XER Technologies

Zurich, Switzerland

Software Engineer

Jan 2026 – Present

- Company-wide AI integration lead: architecting and shipping Claude-based systems across product and internal tooling.
- Shipped an end-to-end flight analysis pipeline: modified mission-planner software to auto-upload custom logs to DB infra, where automated rule-based checks and a Claude-based agent analyse each flight against historical logs and prior reports before the operational team leaves the site. Saves tens of engineer and operator hours weekly across a team of ten.
- Built automated weekly intelligence digests for leadership covering competitor activity, public tender opportunities, and website / traffic performance.
- **Stack:** Claude SDK, MCP, AWS, Supabase, TypeScript, Python.

Machine Learning Intern

Mar 2025 – Dec 2025

- Designed and built XER's core software foundation from scratch: company dashboard (fullstack), Supabase schemas tracking drones, inventory, maintenance, flights and telemetry, auth, server infrastructure (AWS/EC2, later migrated to Railway), and CI/CD.
- Built the operational and field tooling around it: live-telemetry flight display, a utility-controller firmware-flasher tracking every build, drone, operator and timestamp in production, a data plotting tool, and modifications to Mission Planner and QGroundControl.
- **Stack:** Python, Unity/C#, TypeScript, Docker, AWS, Supabase, Railway.

Volvo Cars R&D

Lund, Sweden

Software Intern

Jul 2024 – Aug 2024

- Developed unit and Robot Framework tests in C++ for the Connectivity Team. Flashed and validated release software on test vehicles.

Bosch R&D

Lund, Sweden

Software Intern

Jun 2022 – Sep 2022

- Created an automated Bluetooth testing platform for Android-based In-Vehicle-Infotainment systems (Python, Robot Framework).

SELECTED PROJECTS & AWARDS

Andon Labs Hackathon - 1st place

Jan 2025

- Built an eval using multi-lingual translation chains to degrade MMLU question wording while preserving meaning - probing whether LLMs reason from user intent or pattern-match on training distribution phrasings.

M.Sc. Thesis - Machine Learning for Anti-Poaching

2024-2025

- Decision-tree systems for acoustic gunshot detection, seismic elephant-footstep detection and perimeter intrusion, field-deployed at Sanwild Sanctuary (Limpopo, South Africa).
- Ranger interviews identified false alarms as the primary trust issue, so used Hard Negative Mining and adapted data collection based on real usage feedback. Presented the work as panelist at the MFS National Conference (2026) - on development work in international settings..
- **Stack:** MLFlow, Bayesian Optimisation, Decision Trees, Sequential Feature Selection.

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EDUCATION

Lund University, LTH **Lund, Sweden**
M.Sc. Engineering Physics - Control Theory *Aug 2019 – Jan 2025*

Technical University Berlin **Berlin, Germany**
Erasmus Exchange *Oct 2022 - Aug 2023*

- Coursework: Computer Vision for Remote Sensing, State Estimation for Robotics, Event-Based Robot Vision, Machine Learning 1+2, Robotics and Autonomous Systems. GPA 1.88 (DE)

ESA Academy **ESEC-Galaxia, Belgium**
Selected Student

- ESA/ELGRA Gravity-Related Research Summer School* *Jun 2021*
- Ladybird Guide to Spacecraft Communications Training* *Mar 2023*

TECHNICAL SKILLS

- ML / AI:** Decision trees, scikit-learn, MLFlow, Bayesian optimisation, Hard Negative Mining, Sequential Feature Selection, Computer Vision (YOLO, RANSAC).
- LLMs / Agents:** Claude SDK, Claude Code, MCP servers, RAG with pgvector, agent design, evals, tool-use schema design.
- Programming:** Python, C++, C#, TypeScript, React.
- Infrastructure:** AWS, Supabase, Docker, Railway, CI/CD.
- Robotics / Edge:** ROS1, ArduPilot, MAVLink, NVIDIA Jetson, Mission Planner, QGroundControl, Unity.
- Languages:** Swedish (C2), English (C2), Spanish (C1), Russian (A1).

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Lund Formula Student **Lund, Sweden**
Manager of the Driverless Subteam *Jun 2021 – Sep 2022*

- Led the autonomous subteam (perception with YOLO and RANSAC, sensor fusion, SLAM and path-planning in ROS1/Python/C++). Competed internationally in Hungary and the Czech Republic.

F-Guild **Lund, Sweden**
President of Equality Committee *Jan 2021 - Jan 2022*

- Student safety representative leading a team of six on psychosocial health and equality; coordinated speaker events with LTH equality board.

Earlier: HSSIP Intern at CERN - Geneva (2018), Voluntary Research Assistant, Nanyang Technological University - Singapore (2017).

OTHER ACTIVITIES

Volunteering around the world
Esperanza Verde animal rescue center (Peru 2018); Indómito hostel (Argentina 2023); Yambolombia hostel (Colombia 2024); SanWild Sanctuary (South Africa 2024).

Interests: AI Safety and agent behaviour, conservation technology, programming, intercultural experiences, coffee.